

1 Col AD

CPU or GPU?

A marriage between equals is the one that's likely to thrive and the same holds true for the graphics card. A high-end card without a CPU to match is like a one-horse carriage—it will run, but won't give you the best that it's capable of. At home you can check this for yourself. Here we assume that you have *Quake III* installed and also possess a 3D card in the GeForce range and have a P-III 500 with 64 MB RAM. Start any map in *Quake III* in the multiplayer mode without adding any bots. Bring the console down and type `cg_drawfps1`. Typing this command will show you the average fps on the top

right-hand corner of the screen. Then move around the map, fire rockets and check the frame rate on the top right-hand corner of your screen. After this, press [Esc], add at least five bots and resume the game. You will see that the average fps will be drastically lower, and you will also notice frame drops while playing the game. This means that if you are not into multiplayer gaming and like to play games in single player mode, then you would do fine with a low-end machine but if you want to experience the true power of a high-end 3D card, you should opt for a high-end machine.

aimed at the average gamer who is not looking for steroidal performance and for whom price is a constraint. This card failed in the 3DMark 2001 test but in 3DMark 2000 it gave a score of 3192. The GL Mark score was very low even in the lowest resolution; it returned an average frame rate of a mere 6.1 fps. The card died out in *Quake III* when we ran it at the max settings, logging an average score of 7.5 fps with severe artifacting. We noted the same thing happening again in the *Serious Sam* test and it managed to survive in the low settings mode with an average of 54.9 fps. We could actually physically count the frames in the max settings mode—2.7 fps, to be precise.

Now we move lower on to cards based on the M64 chipset. Looking at the scores of these cards we conclude that they are suitable only for the casual gamer who is not interested in playing games with any frills or at a high

Low-end tests

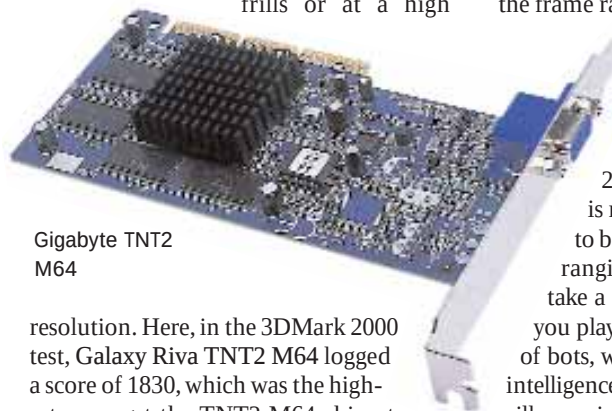
Plain Jane?

The one thing that stood out clearly when we tested the cards on the low-end test bed was that even if you have a powerful card falling in the range of the GeForce2 and higher class, it would not be able to deliver the promised performance on a low-end machine. What this means is that to realise the full potential of a high-end card, you need to have a CPU that complements it. But this doesn't mean that it's impossible to play games on a low-end machine, with acceptable frame rates. Our test logs will prove this point to you.

Take for example the scores logged by the Asus V8200 GeForce3 Deluxe. On our high-end system, it gave phenomenally high scores, but on the low-end system even a card of its calibre couldn't do much to boost the frame rate above a certain point. Here the frame rates in *Quake III* would just not

move above the 60 fps mark in any of the settings—which in itself is quite playable—whereas in the high-end system we reached 209 fps in the normal mode. This

is mainly because the CPU proves to be a bottleneck. Though all cards ranging from GeForce and above take a significant load off the CPU, if you play the game with a large number of bots, which means more AI (artificial intelligence) will have to be calculated, you will experience a severe performance loss. This is because the number crunching, that is making the calculations for each and every bot in the game, is done by the CPU and not by the graphics card.



Gigabyte TNT2
M64

resolution. Here, in the 3DMark 2000 test, Galaxy Riva TNT2 M64 logged a score of 1830, which was the highest amongst the TNT2 M64 chipset based cards in the shootout. Performance was wanting in the GL Mark test where the frame rate logged in the lowest setting was just 7.1 fps.